

Lineage-Aware Hierarchical Deep Learning for Imbalanced Multi-Class Classification of Peripheral Blood Cells

Dissertation Proposal

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Background and the problem

- Reading a blood smear is a core step in diagnosing blood diseases, including leukaemia.
- An expert labels each cell by its shape, chromatin, cytoplasm, and granularity.
- This manual count is slow and repetitive. It also varies between observers.
- It is hardest for rare and immature cells, whose look overlaps with neighbouring types.
- Deep models already match experts on some tasks (Matek et al., 2019).

The problem

Most systems treat all 18 cell types as flat, unrelated labels. They ignore the biology. The data is also highly imbalanced, so rare cells are easily missed.

Research gap and research question

Four research lines are strong on their own, but they have not been combined:

- Deep models for blood-cell images.
- Hierarchical classification that uses label structure.
- Loss functions that are robust to class imbalance.
- Explainable models, such as Grad-CAM.

Research gap

No study so far brings together a biology-based lineage hierarchy, an imbalance-robust loss, and saliency checks on a single, highly imbalanced blood dataset.

Research question

Does a lineage-aware hierarchical classifier, with an imbalance-robust loss, improve fine-grained classification, and especially the rare cell types, compared with a flat classifier on the same data?

Data

- MLL23 dataset: 41,621 single-cell images, 18 classes.
- About a 260-fold gap between the largest and smallest class.
- Two-level label hierarchy: lineage first, then fine cell type.

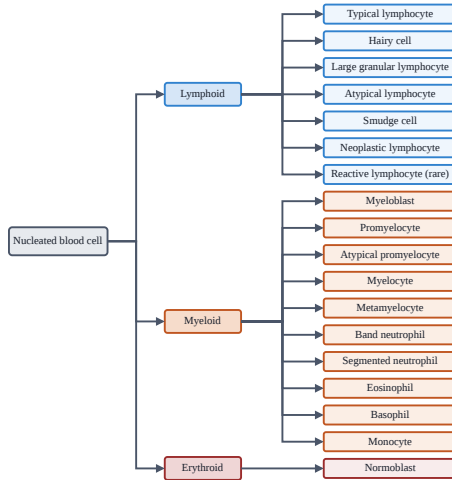
Model

- Shared transfer-learning backbone.
- Backbones to compare: MobileNet, ResNet, ConvNeXt, and a Vision Transformer.
- Two heads: one for lineage, one for fine type.

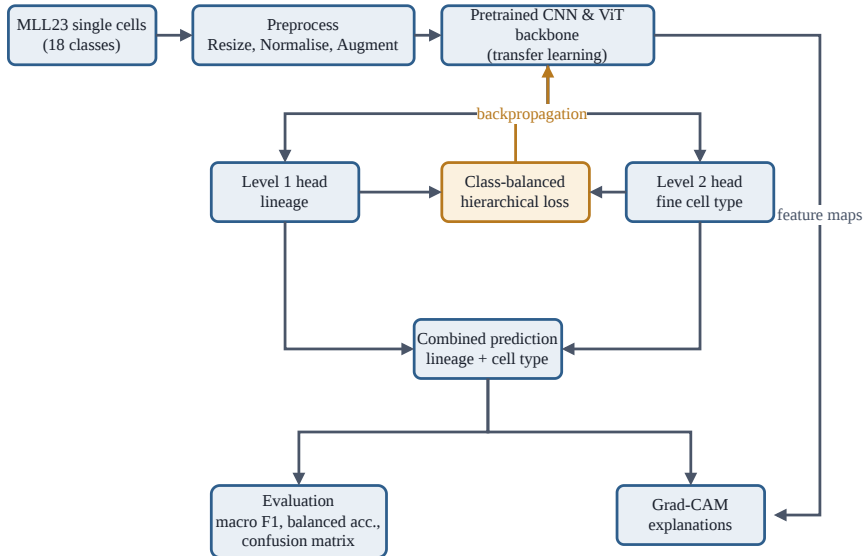
Training and evaluation

- Stratified split, so rare classes appear in every set.
- Stronger augmentation on minority classes.
- Compare against a flat baseline.
- Ablation: remove the loss term, then the hierarchy.
- Metrics: macro F1, balanced accuracy, confusion matrix.
- Grad-CAM for visual explanations.

Two-level lineage hierarchy



Proposed pipeline



Expected outcomes

- Higher macro F1 and balanced accuracy than the flat baseline.
- The largest gains on the rare cell types.
- Fewer errors that cross unrelated lineages.
- Ablation shows what the hierarchy and the loss each add.
- Grad-CAM confirms the model looks at meaningful regions, such as the nucleus, the chromatin, and the cytoplasm.
- A reusable, interpretable pipeline for imbalanced single-cell classification.

- The model is shaped by blood-cell biology, not by flat labels.
- It joins a lineage hierarchy, an imbalance-robust loss, and saliency checks in one pipeline.
- It is tested on the largest public single-cell blood dataset.
- The focus is the rare cells that matter most in the clinic.

Thank you. Questions are welcome.